

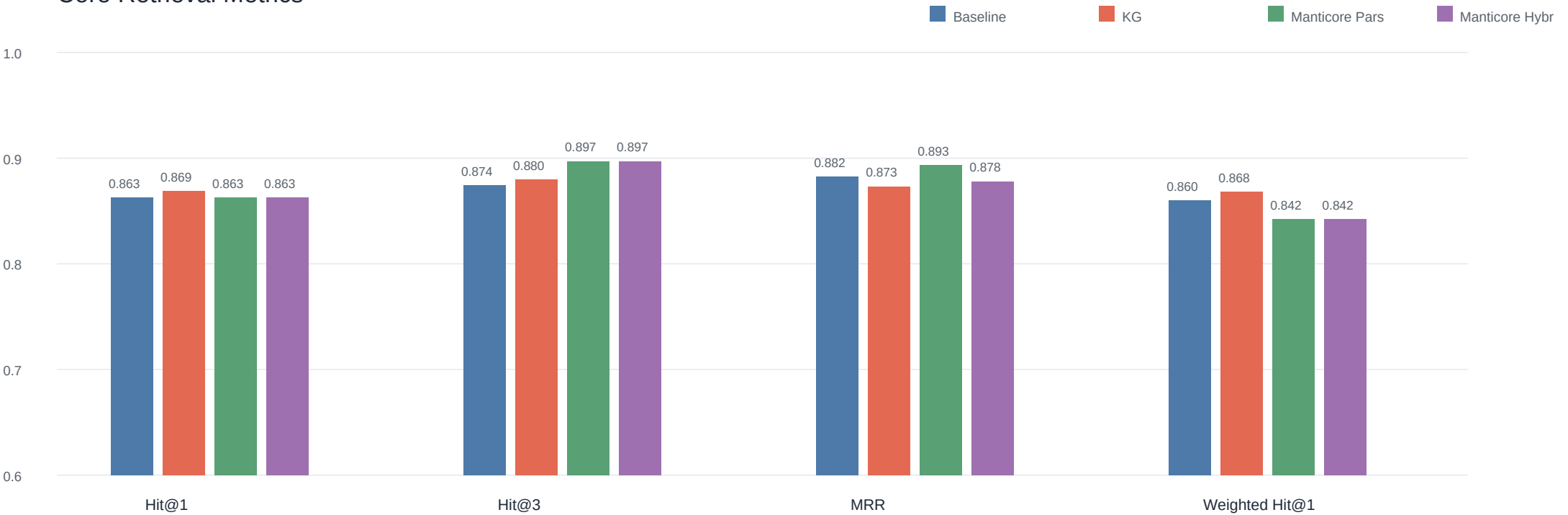
RTL Retrieval Methods: Performance Comparison

Frontend updated: methods 1-3 now use tree-sitter-verilog generated seed.

Dataset: 175 questions, 1012 indexed modules

Method	Hit@1	Hit@3	MRR	Weighted Hit@1	Avg query ms
Baseline	0.8629	0.8743	0.8824	0.8601	39.480
KG	0.8686	0.8800	0.8733	0.8683	49.629
Manticore Parser/LSP	0.8629	0.8971	0.8935	0.8423	46.801
Manticore Hybrid	0.8629	0.8971	0.8781	0.8423	53.774

Core Retrieval Metrics



Method Characteristics

All four methods are now comparable around a tree-sitter frontend, except Graphify keeps its own AST graph model.

Method	Input signals	Ranking	Strength	Tradeoff
Baseline	tree-sitter seed: module name, file path, ports, instances	Parser/LSP-style lexical overlap with exact anchors	Fast simple lookup on structural RTL fields	No ontology-only graph reasoning
KG	tree-sitter seed plus labels, summaries, reverse parents	KG-aware scorer with semantic expansion	Best Hit@1 in this tree-sitter run	Depends on label quality and seed coverage
Manticore Parser/LSP	tree-sitter parser/LSP fields only	Manticore-style BM25F full-text field weighting	Full-text ranking without KG dependency	No KG-only relationship context
Manticore Hybrid	tree-sitter parser/LSP fields plus KG labels/summaries/parents	BM25F with KG field boosts	Combines text ranker and KG fields	Slowest local proxy query latency

Hit@1 by Difficulty Level

Method	L1	L2	L3	L4	L5
Baseline	0.8000	1.0000	0.8571	0.7714	0.8857
KG	0.8000	1.0000	0.8571	0.7714	0.9143
Manticore Parser/LSP	1.0000	1.0000	0.7429	0.7143	0.8571
Manticore Hybrid	1.0000	1.0000	0.7429	0.7143	0.8571

Reading: KG has the best Hit@1 in this tree-sitter run; all methods now share a stronger structural seed.
Manticore remains a local BM25F proxy and writes load-ready schema/documents for a real server-backed run.